Introduction to Python Programming

Objectives

- Familiarize writing codes using Python as the programming language.
- Write programs in Python.

Important items to remember in Python Programming

Creating variables

- Python has no command for declaring a variable.
- A variable is created the moment you first assign a value to it.

```
x = 5
y = "Juan dela Cruz"
print(x)
print(y)
```

```
5
Juan dela Cruz
```

Creating variables

- Variables do not need to be declared with any particular *type and* can even change type after they have been set.

```
#Example 13
x = 25        # the data type of x is integer
print(x)
x = "Juan dela Cruz" # the data type of x is now string
print(x)
```

```
25
Juan dela Cruz
```

Casting

- If you want to specify the data type of a variable, this can be done with casting.

```
#Example 14
x = str(3)     # x will be '3'
print (x + '1')

y = int(3)     # y will be 3
print (y+1)

z = float(3)   # z will be 3.0
print (z+0)
```

```
31
4
3.0
```

Get the Type

- You can get the data type of a variable with the `type()` function.

```
#Example 15 (Get the type)
x = 8
y = "Juan Dela Cruz"
print(type(x))
print(type(y))
```

```
<class 'int'>
<class 'str'>
```

Single or Double Quotes?

- String variables can be declared either by using single or double quotes:

```
#Example 16
x = "Juan dela Cruz"
# is the same as
x = 'Juan dela Cruz'
```

Case-Sensitive

- Variable names are case-sensitive.

The code below will create two variables.

```
#Example 17
var1 = 2
Var1 = "Pedro"
#Var1 will not overwrite var1

print(var1)
print(Var1)
```

```
2
Pedro
```

Variable Names

There are several techniques you can use to make them more readable

- Camel Case

Each word, except the first, starts with a capital letter

- Pascal Case

Each word starts with a capital letter

- Snake Case

Each word is separated by an underscore character

Variables- Sample Code

```
#Example 18
#CamelCase
myVariableName = "Pedro"

#Pascal Case
MyVariableName = "Pedro"

#Snake Case
my_variable_name = "Pedro"
```

Many Values to Multiple Variables

- Python allows you to assign values to multiple variables in one line:

```
#Example 19
x, y, z = "Orange", "Banana", "Cherry"
print(x)
print(y)
print(z)
```

```
Orange
Banana
Cherry
```

One Value to Multiple Variables

- And you can assign the *same* value to multiple variables in one line.

```
#Example 20
x = y = z = "Orange"
print(x)
print(y)
print(z)
```

```
Orange
Orange
Orange
```

Unpack a collection

- If you have a collection of values in a list, tuple etc. Python allows you to extract the values into variables. This is called unpacking.

```
#Example 21
fruits = ["apple", "banana", "cherry"]
x, y, z = fruits
print(x)
print(y)
print(z)
```

```
apple
banana
cherry
```

Output Variables

- The Python print() function is often used to output variables.
- In the print() function, you output multiple variables, separated by a comma:

```
#Example 22
x = "Python is awesome"
print(x)

a = "Python"
b = "is"
c = "awesome"
print(a, b, c)
```

```
Python is awesome
Python is awesome
```

Output Variables

- You can also use the + operator to output multiple variables:

```
a = "Python "
b = "is "
c = "awesome"
print(a + b + c)
```

```
Python is awesome
```

- Notice the space character after "Python " and "is ", without them the result would be "Pythonisawesome".

Output variables

- For numbers, the + character works as a mathematical operator:

```
#Example 23
x = 20
y = 5
print(x + y)
```

```
25
```

- In the print() function, when you try to combine a string and a number with the + operator, Python will give you an error:

Output Variables

- The best way to output multiple variables in the print() function is to separate them with commas, which even support different data types:

```
#Example 24
x = 5
y = "John"
print(x, y)
```

```
5 John
```

Practice Problem 1

1. Create Python program that accepts an input integer from the user and displays the equivalent conversion and grade description. Refer to the table below.

Total Grade	Conversion	Grade Description	
96-100	4.0	A	High Distinction
90-95	3.5	B+	Distinction
85-89	3.0	B-	Very Good
80-84	2.5	C+	Good
75-79	2.0	C-	Average
<75	1.0	F	Fail

Practice Problem 2

2. Create a Python program that accepts three input numbers from the user and determines the highest and the smallest numbers based on the inputted value.